

PathFlowQuery-GNN: A Spatio-Temporal Learning Approach for Urban Freight Path Flow Estimation

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1. Introduction

Background

- Reliable freight traffic demand estimation is essential for fine-grained freight management.
- link-level counts and aggregate indicators are insufficient to reveal time-varying OD demand, route choice, and path-level traffic patterns.

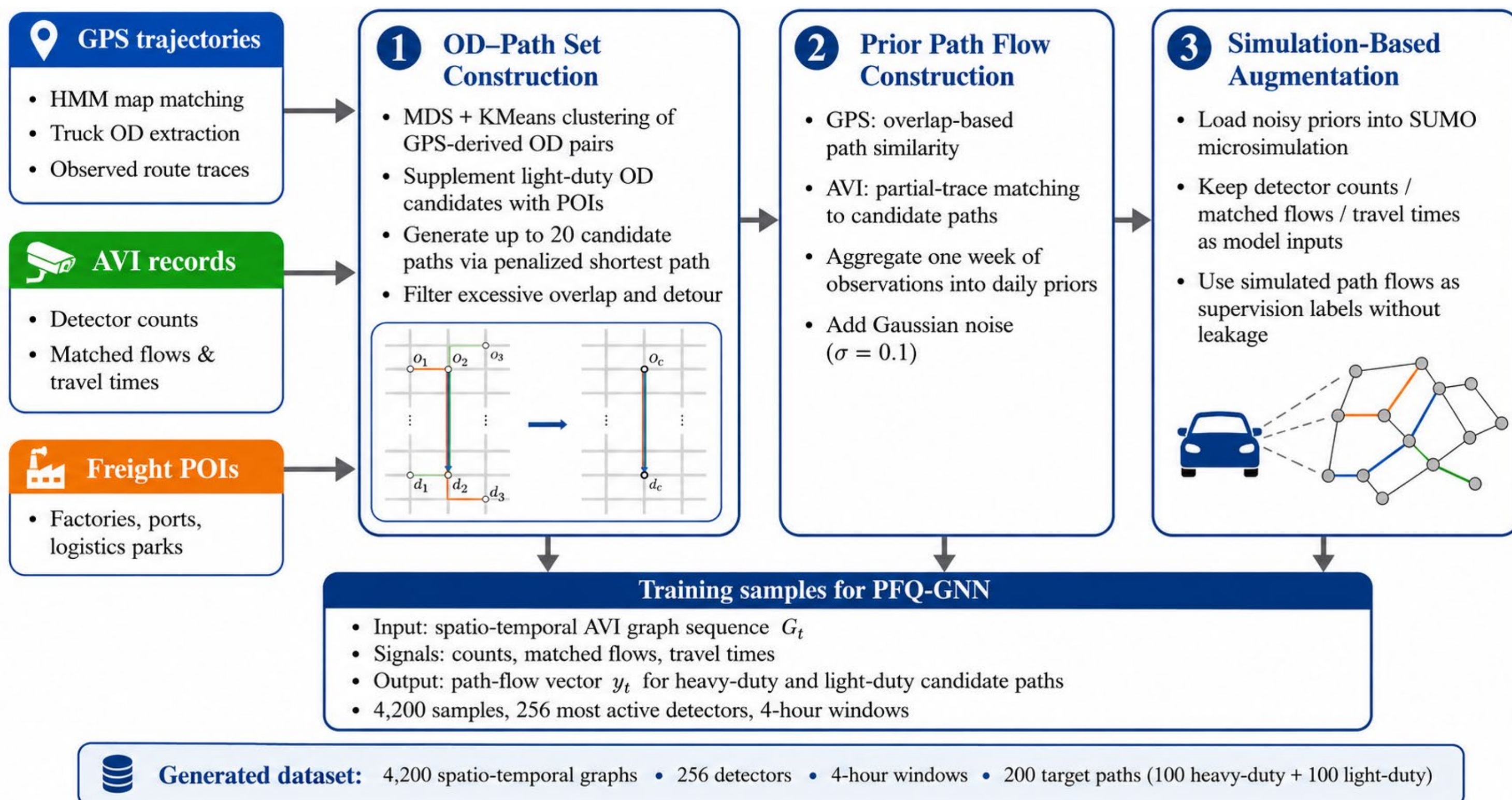
Challenge

- This study aims to estimate time-varying urban freight path flows directly from AVI detector streams. The key challenge is to infer high-dimensional path-level flows from sparse, localized detector observations.

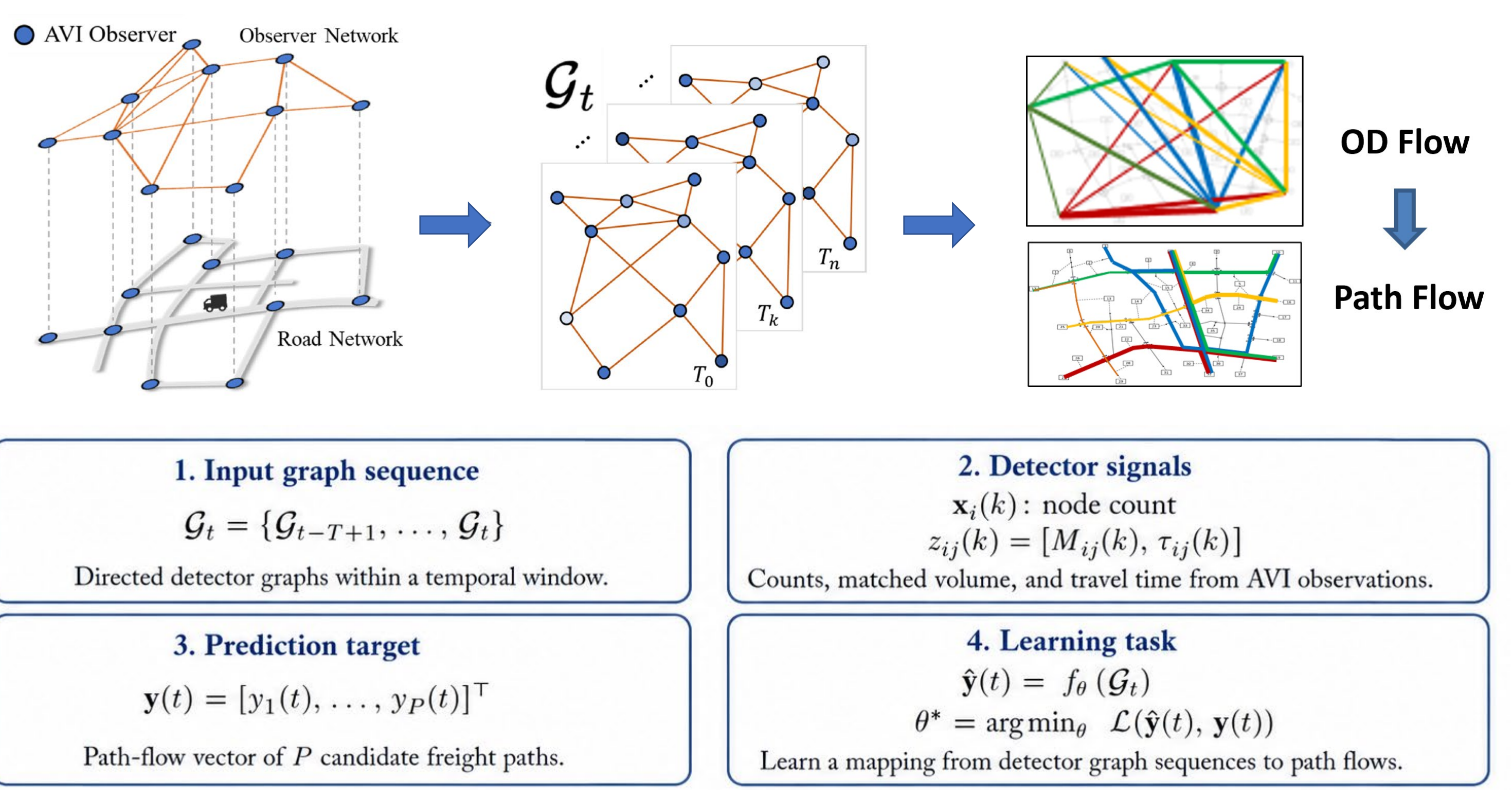
Contributions

- This study proposes a city-scale, dynamically adaptive framework for path-level freight flow estimation that integrates data fusion, simulation-based augmentation, and spatio-temporal graph learning framework PFQ-GNN.

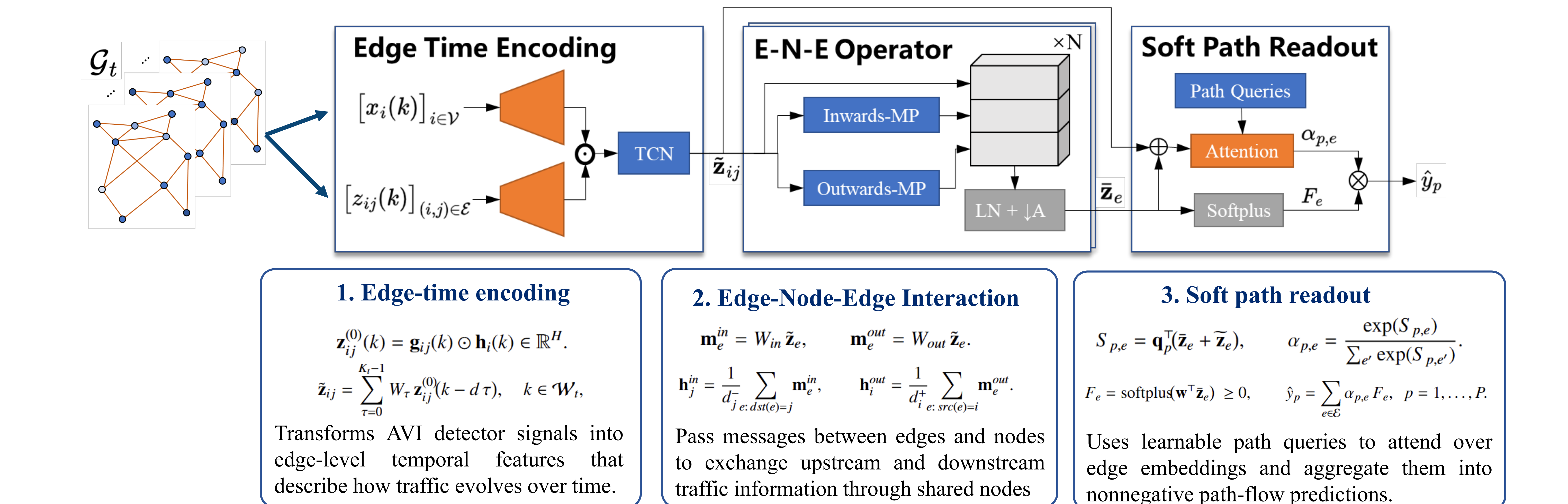
2. Data Dre-Processing



3. Model Formulation



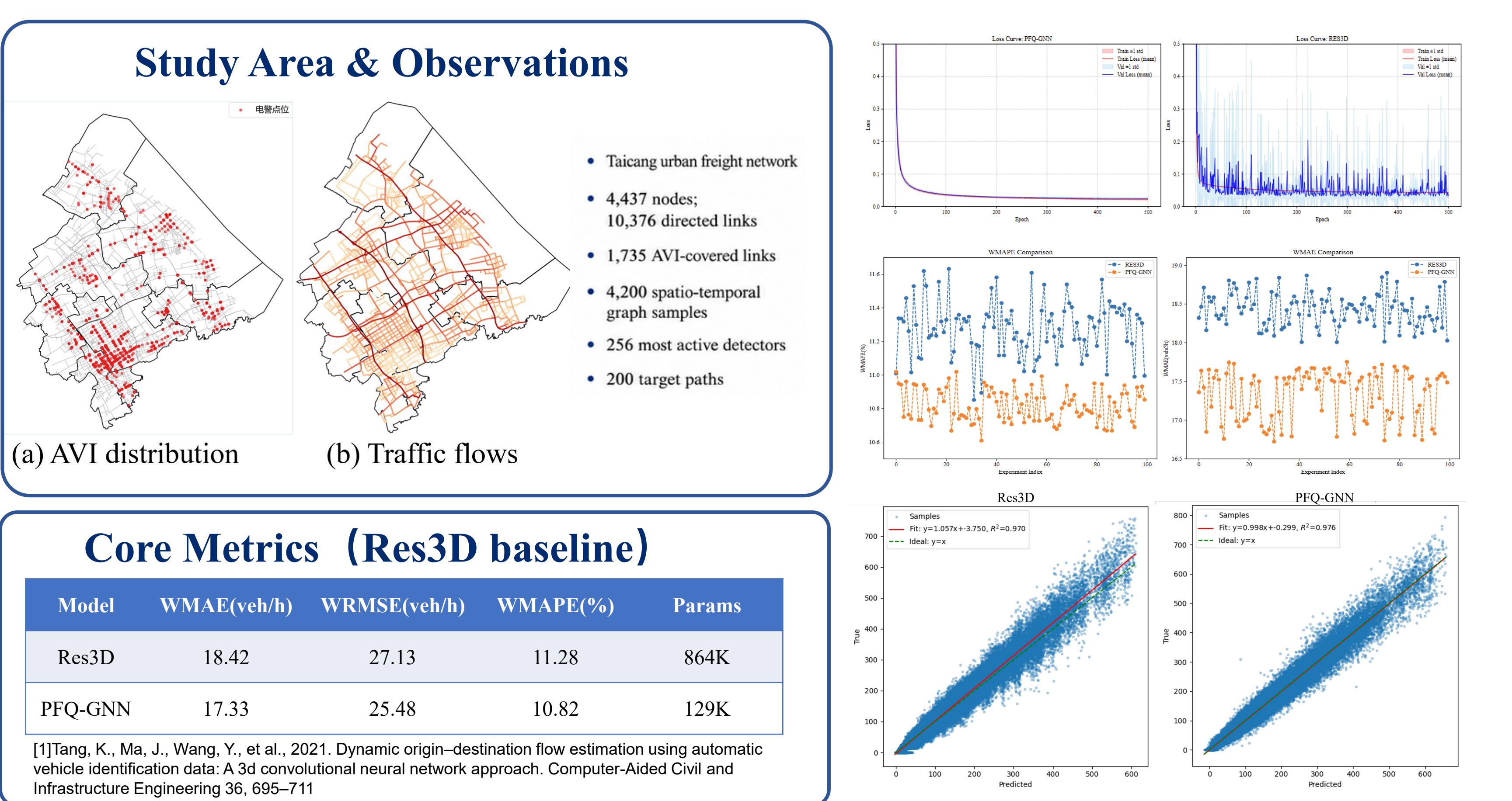
4. PFQ-GNN Architecture (Our Proposed Framework)



Loss with Regularization: Standard MSE + Attention entropy + Smooth path

$$\mathcal{L} = \frac{1}{P} \sum_{p=1}^P (\hat{y}_p(t) - y_p(t))^2 + \lambda_{\text{ent}} \frac{1}{P} \sum_{p=1}^P \text{KL}(\alpha_p \| \mathbf{u}) + \lambda_{\text{sm}} \frac{1}{P} \sum_{p=1}^P \alpha_p^\top L_e \alpha_p$$

5. Experimental Results



6. Conclusion

- PFQ-GNN provides a compact and interpretable spatio-temporal graph learning framework that directly maps AVI detector streams to path-level urban freight flows.
- By integrating edge-time encoding, E→N→E interaction, and regularized soft-path readout, the model achieves better accuracy, stability, and parameter efficiency than the Res3D baseline.
- The learned soft-path attention reveals corridor-level freight movement patterns, supporting evidence-based freight corridor management and future multimodal traffic data fusion.
- Future Work: (i) couple learning with differentiable assignment and systematic back-validation to independent AVI aggregates, (ii) develop uncertainty-aware for risk-sensitive operations,